TOREL SAMYUKTHA S

Int BS-MS Earth & Environmental Sciences, Geology Major | IISER Bhopal, Madhya Pradesh - 462066
Email | LinkedIn | Github | Portfolio

SUMMARY

A MS geology graduate student specialised in CO_2 sequestration in basaltic formations. Developed and validated a 1D reactive transport model (Python-Phreeqc 3.8.6) to simulate CO_2 mineralisation, porewater chemistry evolution, porosity -permeability coupling and injection strategy optimisation in Deccan basalts. Additional expertise in XRD, seismic data analysis and python based machine learning.

EDUCATION

Indian Institute of Science Education and Research, Bhopal *Dec 2021- May 2026*
Int BS-MS Earth & Environmental Sciences, Geology major CGPA: 8.23/10
Coursework: Mineralogy, Petrology, Sedimentology, Geophysics, Geology of fuels, Remote sensing, Seismology

MS THESIS

Reactive Transport Model on CO_2 Mineralization in Malwa Plateau Basalts (Deccan Traps):
Implications of Basalt Reactivity and injection strategies *Aug 2025 - May 2026*
IISER Bhopal | MS thesis | PGSG lab | Dr. Jyotirmoy Mallik

- Built a 1D Python-PHREEQC reactive transport model simulating CO_2 mineralisation, porewater chemistry, porosity-permeability evolution (Kozeny-Carman) and mineral kinetics (TST rate law) in Deccan basalts.
- Compared three injection strategies (Single-burst, continuous and pulsed WAG) across Mhow and CRBG basalt systems validated against Wallula pilot projects and CRBg laboratory datasets.
- Field documentation of 20 outcrops; XRD, petrographical and petrophysical characterisation of lava flow sequences across Malwa plateau.

PUBLICATIONS

1. **Samyuktha, T**., Adak, S., & Mallik, J. (2026). Evaluating Single-burst, Continuous and WAG CO_2 injection through reactive transport modelling in Basalts: Controls on Carbonation and Reservoir Evolution.
(manuscript submitted)
2. Babu, A.S., **Samyuktha, T.,** Mallik, J.*. "GeoRTM: Rapid Screening of CO_2 Mineralization Potential in Mafic and Ultramafic Systems using Reactive Transport Modelling." (manuscript submitted)

CONFERENCE ABSTRACT

SEG Annual Meeting (Abstract Accepted) *June* 2026
"Prospective CO_2 injection horizons within Malwa Plateau basalts, Deccan Volcanic Province" - preliminary results on basalts' reactivity and mineral fluid dynamics relevant to in situ CO2-mineralisation.

TECHNICAL SKILLS

Modelling: PHREEQC 3.8.6, IPHREEQC (python), 1D reactive transport, TST kinetics, Kozeny-Carman permeability
Programming: Python (NumPy, Pandas, Matplotlib, Scipy, Scikit-learn), MATLAB, R
Analytical: XRD, petrographic thin section analysis, porosity measurement
Geospatial & Seismic: QGIS, Google Earth Engine, SplitLab 1.9.0, SAC, GMT, STA/LTA earthquake detection

GUEST PRESENTATION

KUAS | Kyoto, Japan *Jul 2025*
During my internship at Hiyoshi, I had been provided with an opportunity to give a presentation on "Detection of mantle deformation using seismic anisotropy"

INDUSTRIAL INTERNSHIP

Hiyoshi Corporation, Japan *May 2025 - Jul 2025*
Gained knowledge in water and food quality analysis through hands-on experience on analysis (organic, inorganic, pesticides, aflatoxins) using advanced technologies. Had a good understanding of different water treatment plants & pumping stations and market on PFAS analysis in India.
Skills: Environmental chemistry analysis, Market research

RESEARCH INTERNSHIP

Detection of Seismic anisotropy: Shear wave splitting: *Jun 2024 - Aug 2024*

Dr. Ashwani Kant Tiwari | IISER Bhopal
Processed existing seismic datasets & extended analysis to real data to interpret the splitting parameters to understand seismic anisotropy.
Skills: Splitlab 1.9.0, Seismic data analysis

Introduction & Applications of Remote Sensing, GIS & Drone Technology: *Dec 2023 - Jan 2024*
Geovigyan | Dr. Gaurav Singh | Remote
Gained hands-on experience in GIS applications using QGIS. Utilized google maps & google earth engine for geospatial visualization & data interpretation.
Skills: QGIS, Google Earth Engine

Pre-processing of Earthquake Data and Automatic Detection of Earthquake using python:
Dr. Ashwani Kant Tiwari | IISER Bhopal *May 2023 - Jul 2023*
Designed & Automated a python based work flow for analyzing seismic data from IRIS. Enhanced data quality & accuracy by applying digital filters, also implemented STA/LTA algorithm to enable real-time earthquake detection.
Skills: Anaconda, Python, Seismic data handling

WORKSHOP

Fundamentals of petroleum geology and its application in the oil & gas industry: *March 2026*
IISER Bhopal | PGSG lab
Participated in a 2 day technical workshop on petroleum systems covering source rock geochemistry, reservoir and cap rocks,petrophysics, seismic exploration methods with well log interpretation.
Skills: Business challenge, Seismic exploration

Carbon Capture & Storage: From concepts to field implementations:
IISER Bhopal | PGSG lab *Oct 2025*
Participated in a 3 day workshop on Carbon Capture and Storage, implying major CO2 storage (Depleted oil & gas, Coal bed methane, Saline aquifers, Basalt formations) and the status of CCS in India. Included a field visit to basalt outcrop to observe red boles, vesicular and massive basalts.
Skills: Field Observation, Basalt Geochemistry, Reactive Transport Model

EarthScope Consortium Seismology Skill Building Workshop 2024:
Mike Brudzinski (Miami University) | Gillian Haberli (EarthScope Consortium) *Jun 2024 – Sep 2024*
Completed a 70-hour online course in scientific computing and computational thinking skills while working with seismic data. Executed a project on the seismic data analyses on the Bhuj Earthquake that occurred in 2001 in Gujarat, India. Project link
Skills: Linux, Python, GMT, SAC, Seismic data analyses

PROJECT

Sales Prediction & Analysis:
Dr. Samiran Das | IISER Bhopal *Nov 2024*
Developed and evaluated machine learning models (Linear Regression, KNN Regressor, Decision Tree, Random Forest, Gradient Boosting) to forecast future sales. Identified Gradient Boosting Regressor as the most effective for reliable sales prediction.
Skills: Python, Data processing, feature engineering, regression modeling, predictive analysis

ADDITIONAL CERTIFICATION COURSES

- Reservoir Geomechanics
- Introduction to Python
- MATLAB Programming
- MySQL
- Remote Sensing & GIS in Natural Resource management (ISRO)

POSITION OF RESPONSIBILITY

Held multiple organisational and leadership roles within the institute, including serving as the Departmental Representative for Earth and Environmental Sciences and as a Core Committee Member for the Carbon Capture & Storage Workshop. Contributed to academic outreach and student engagement through the EES Club and the Humans of IISERB editorial team. Coordinated Continuum 2023 (Math Fest) and supported various departmental and community events, including EES Meet, MAHUA, and Geovigyan, developing strong event-management, communication, and collaborative skills.

LANGUAGES

Tamil (Native), English (Proficient), German (A2), Hindi (Conversational), Malayalam (Conversational)